Input files

Technology related

(Foundries/PDK Team)

1. Technology file (.tf) (✓)
2. Liberty file / Library file (.lib) (✓)
3. Library exchange format (.lef) (✓)
   Layout exchange format
4. tlut file (✓)
   (table lookup files)

Design Related.

1. Netlist (.v/.vhd) (✓)
2. MCMM → multi corner multi mode (SDC)
3. UPF → Unified power format file (.upf)
4. Scandef

① Library/Liberty file. (.lib)

.lib → human readable format → IEEE standard

.db → binary format → Synopsys tools
Runtime is less.

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nldm (non-linear delay modelling)	CCS (Composite Current Source)
① MOSFET is represented voltage source	① MOSFET is represented as Current source
② Less accurate	② More accurate
③ Less runtime	③ More runtime

Content of Liberty files

① Units for parameters like time, current, Voltage, power, capacitance.

② Wire load Model table → To calculate RC values before placement stage

↳ It used to estimate net length based on fanout before placement stage.

Netlength → 30 μm

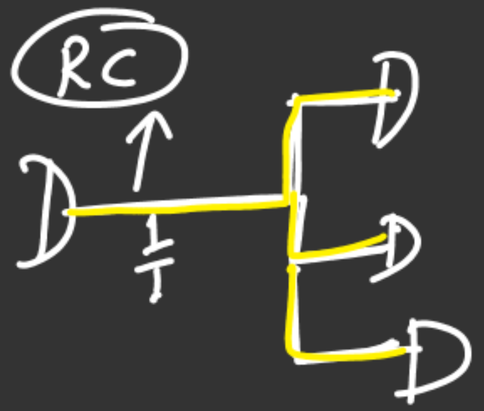
$C = C_\mu \times \text{Netlength}$

$= 0.29\ \text{fF}/\mu\text{m} \times 30\ \mu\text{m}$

$= 8.7\ \text{fF}$

$R = R_\mu \times \text{Netlength}$

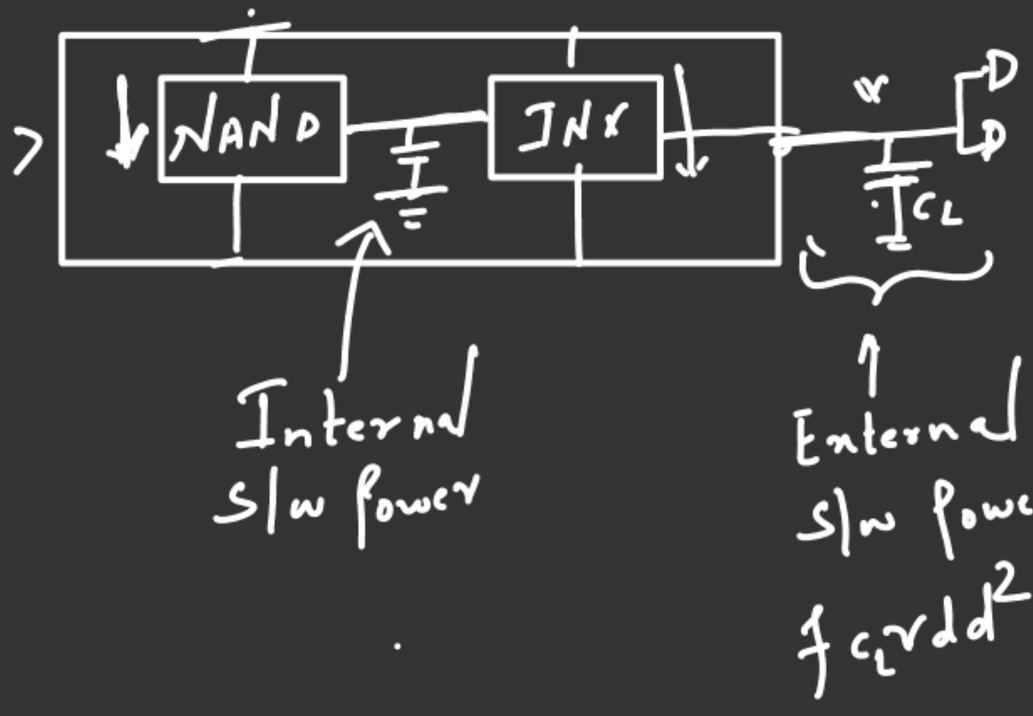
$= 2.27 \times 10^{3}\ \Omega/\mu\text{m} \times 30\ \mu\text{m} = 68.1\ \Omega$



fanout → 3

For Every Cell.

- Area of Cell
- Leakage Power information
- Internal power table → Power dissipation within cell
  - Short ckt Power + Inter s/w Power.
  - depends on i/p transition & o/p Load
- o/p transition table
  - i/p transition and o/p Load
- Cell delay table
  - i/p transition and o/p Load.
- Setup & hold tables for flip-flops.

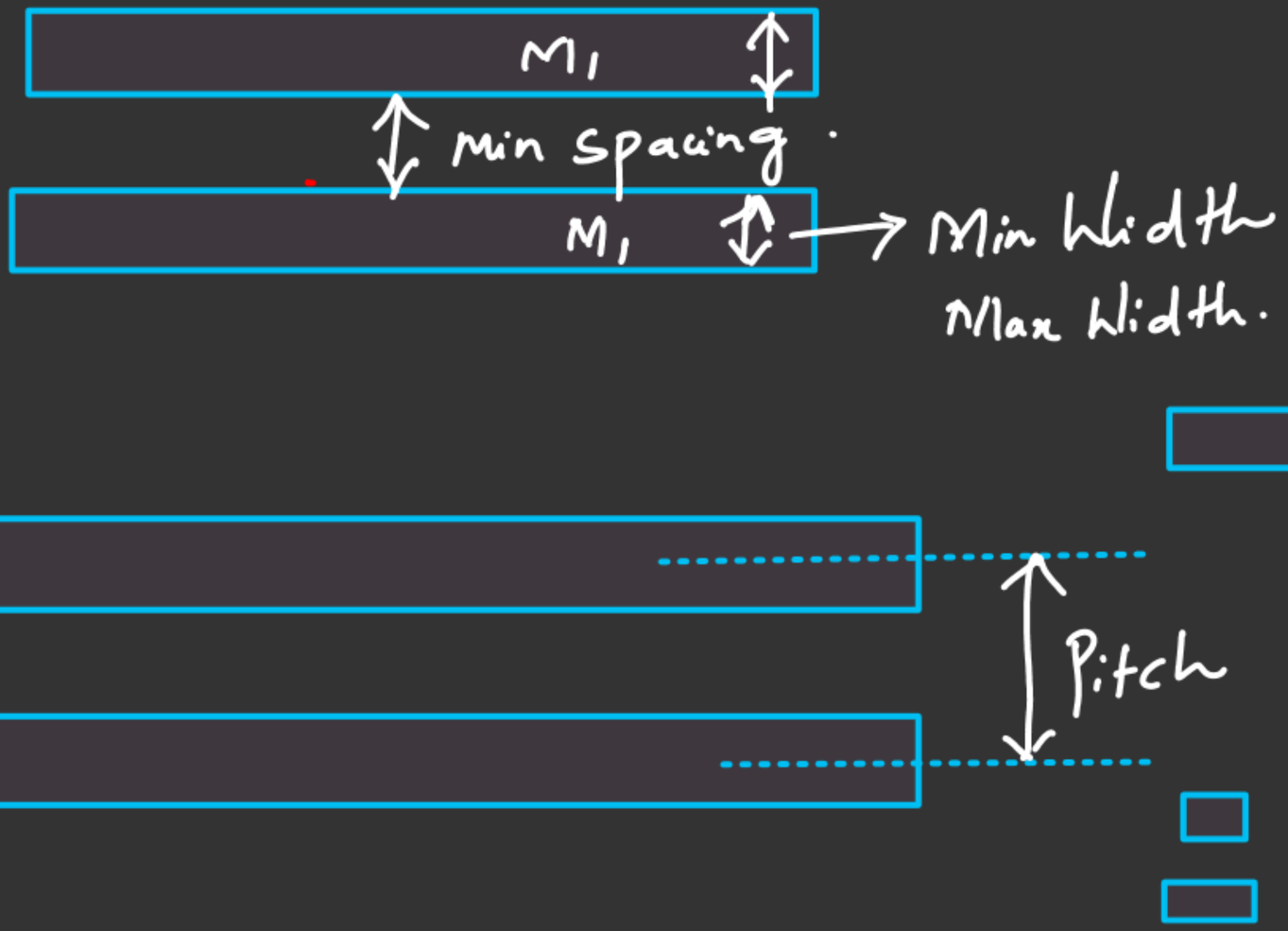


② Technology file (.tf) / tech_def

- Synopsys
- Cadence.

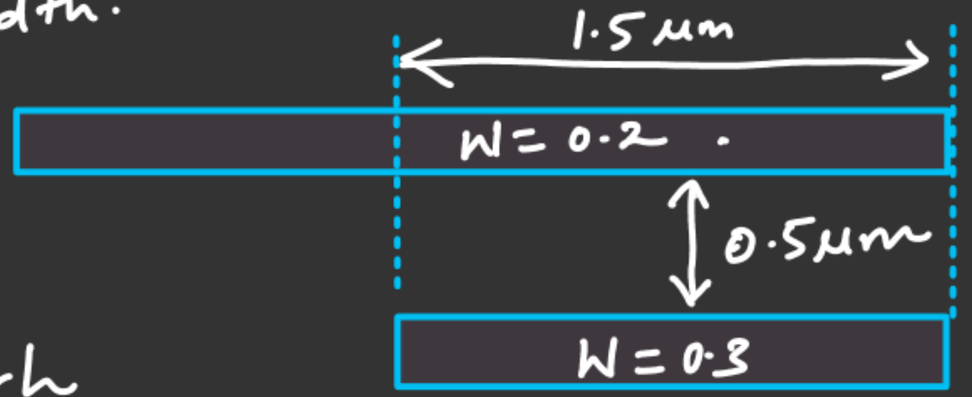
↳ Units for parameters
↳ Site rows & Site Colums information.
↳ Rules for Metal Layers.
  ↳ Min width
  ↳ min spacing
  ↳ Spacing table based on width and parallel length
  ↳ pitch of metal layer.
↳ Rules for Vias
  ↳ Minimum enclosure distance
  ↳ Cut to Cut Spacing. min Cut height min Cut Width.
↳ Metal density rules
↳ Current density table for metal layers

Metal layer Rules.



↳ Spacing table based

Metal Width & parallel length



Site row heigh & Site Coloumn Width.

Height of std_cell

↓

$h = n \times H$

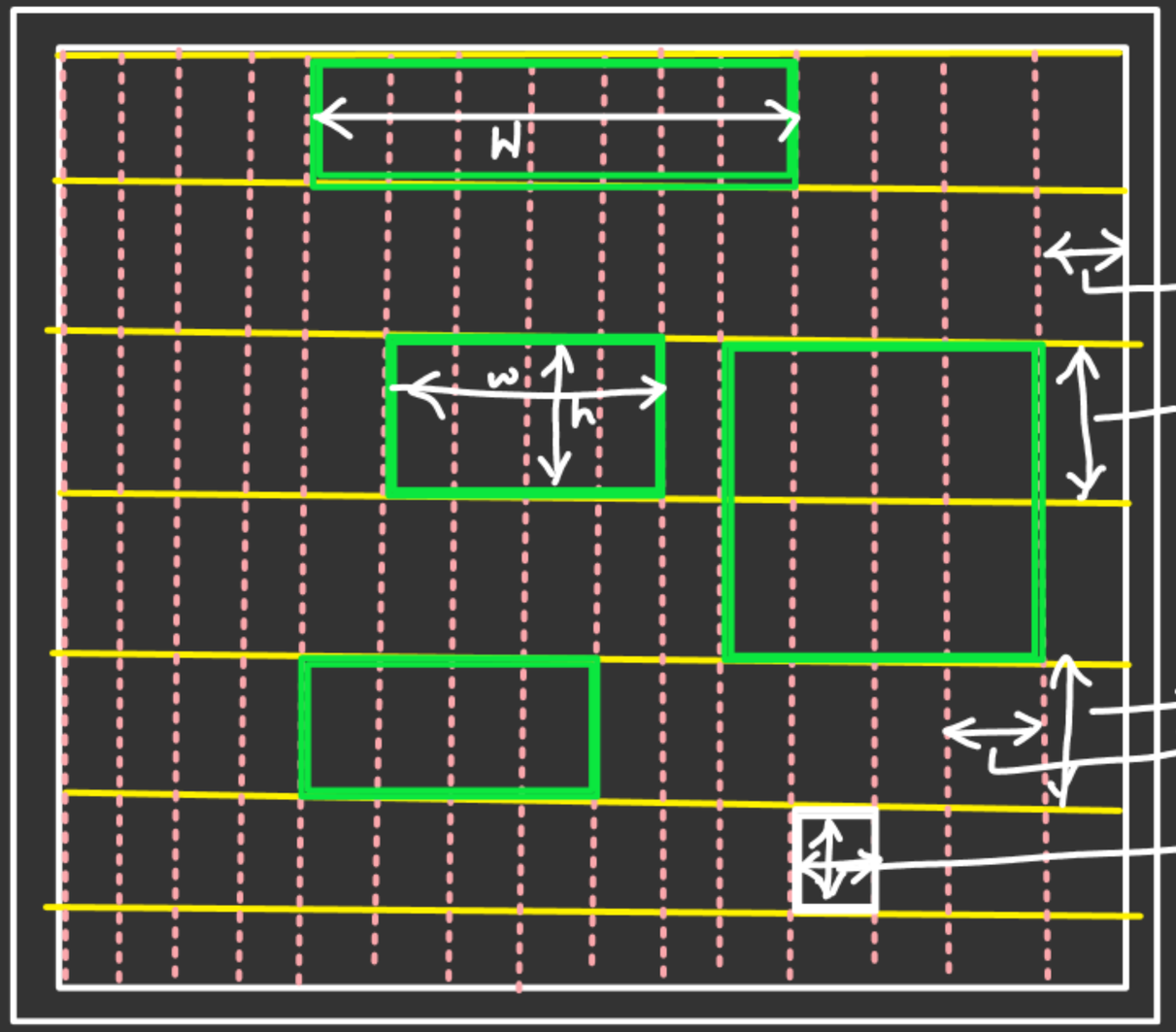
$n \rightarrow$ integer, generally $n = 1$

$w = m \times W$

$m \rightarrow$ Integer

Tile height → Site row height

Tile Width → Site coloumn Width.



W → Site Coloumn width

H → Site row height

1.672 μm

0.152 μm

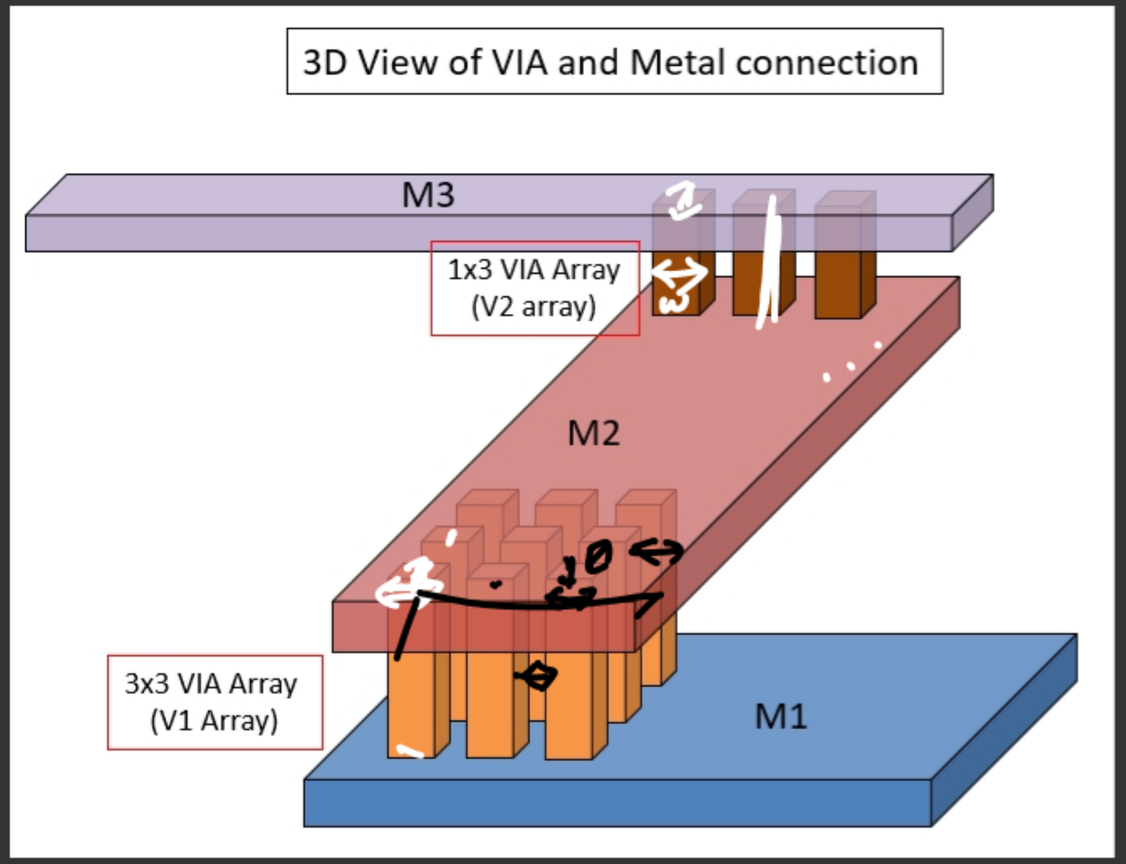
Tile

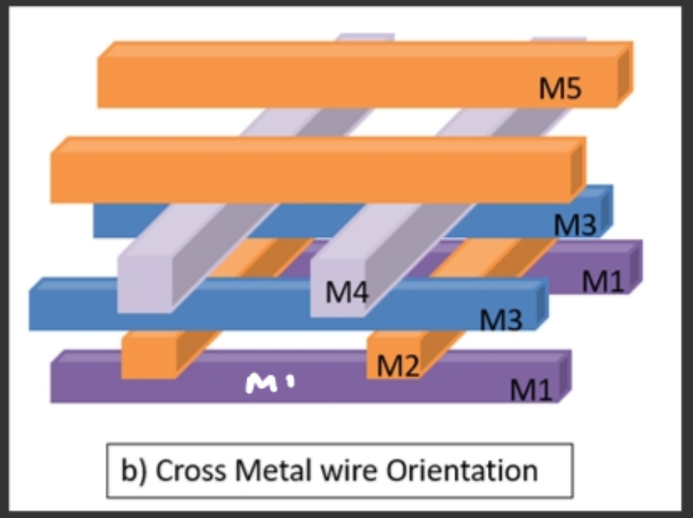
Metal layers.

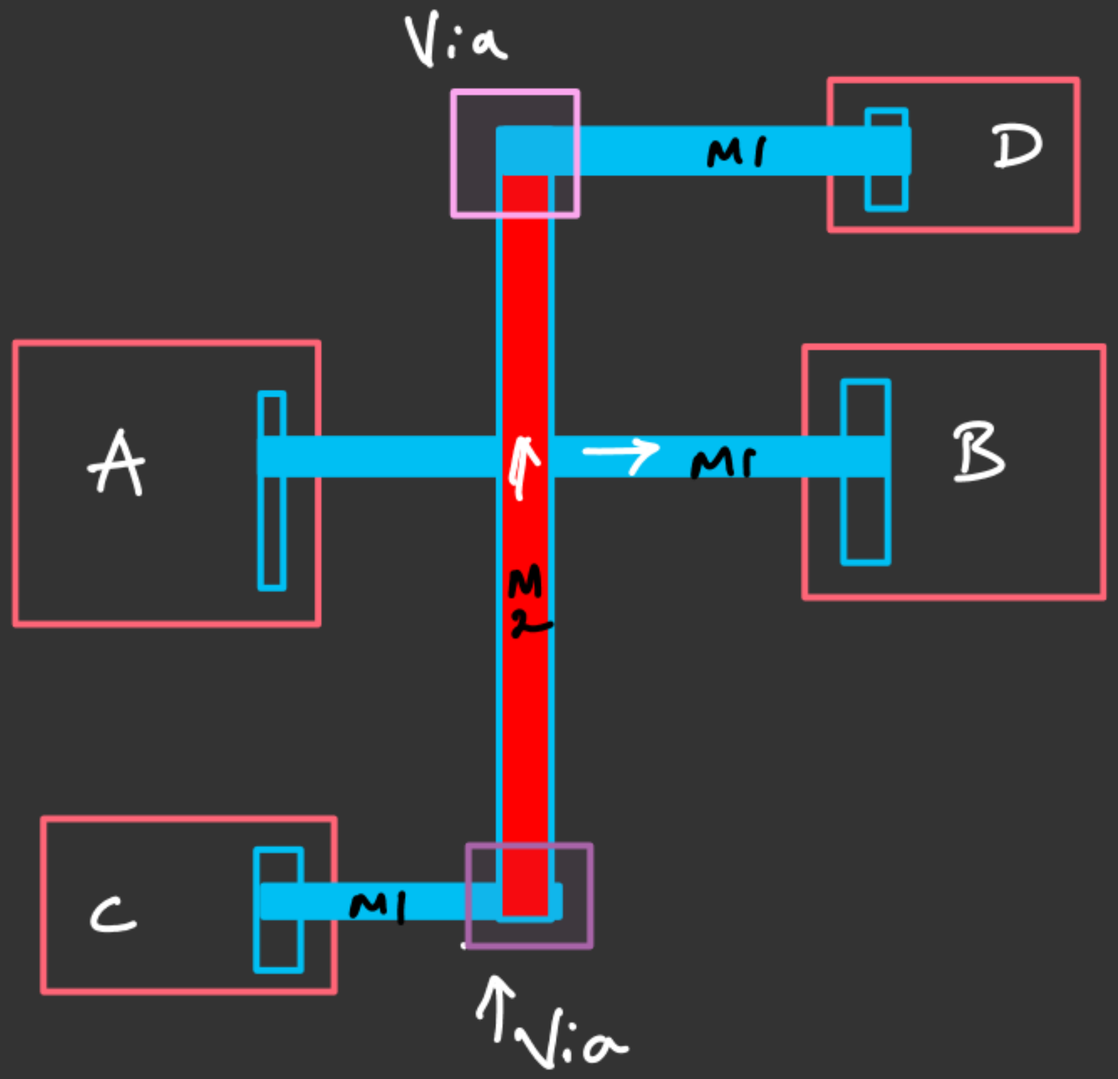
M1 → Horizontal	M5 → H	M9 → H
M2 → Vertical	M6 → V	
M3 → Horizontal	M7 → H	
M4 → Vertical	M8 → V	

Vias → Are used to connect one metal layer to another metal layer in different plane

Cuts → Number of connections inside Via is called as Cuts


3D View of VIA and Metal connection
M3
1x3 VIA Array
(V2 array)
M2
3x3 VIA Array
(V1 Array)
M1


M5
M3
M4
M1
M3
M2
M1
b) Cross Metal wire Orientation


Via
M1
D
A
M1
B
M2
C
M1
Via

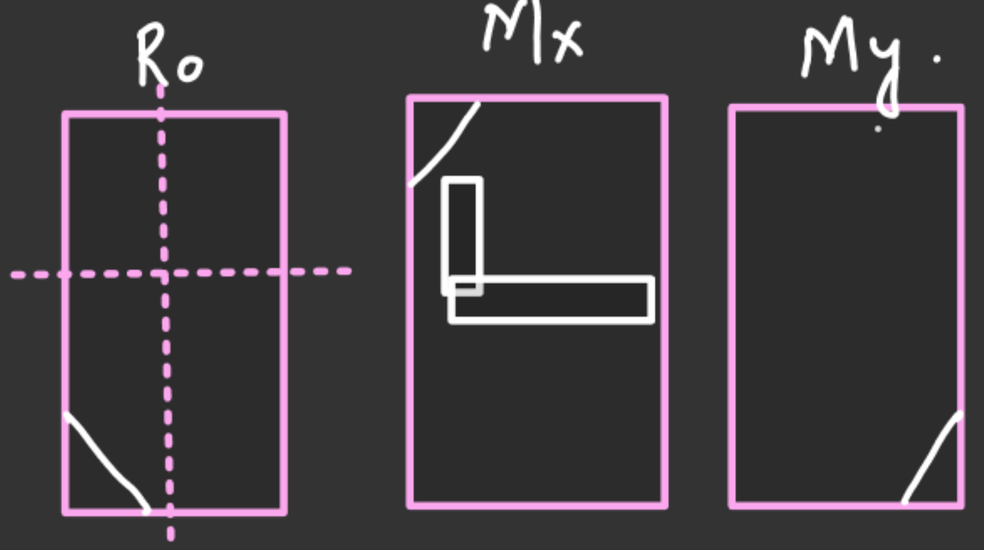
③ Layout Exchange format (LEF)
Library Exchange format

↳ It has abstract layout information of standard cells.

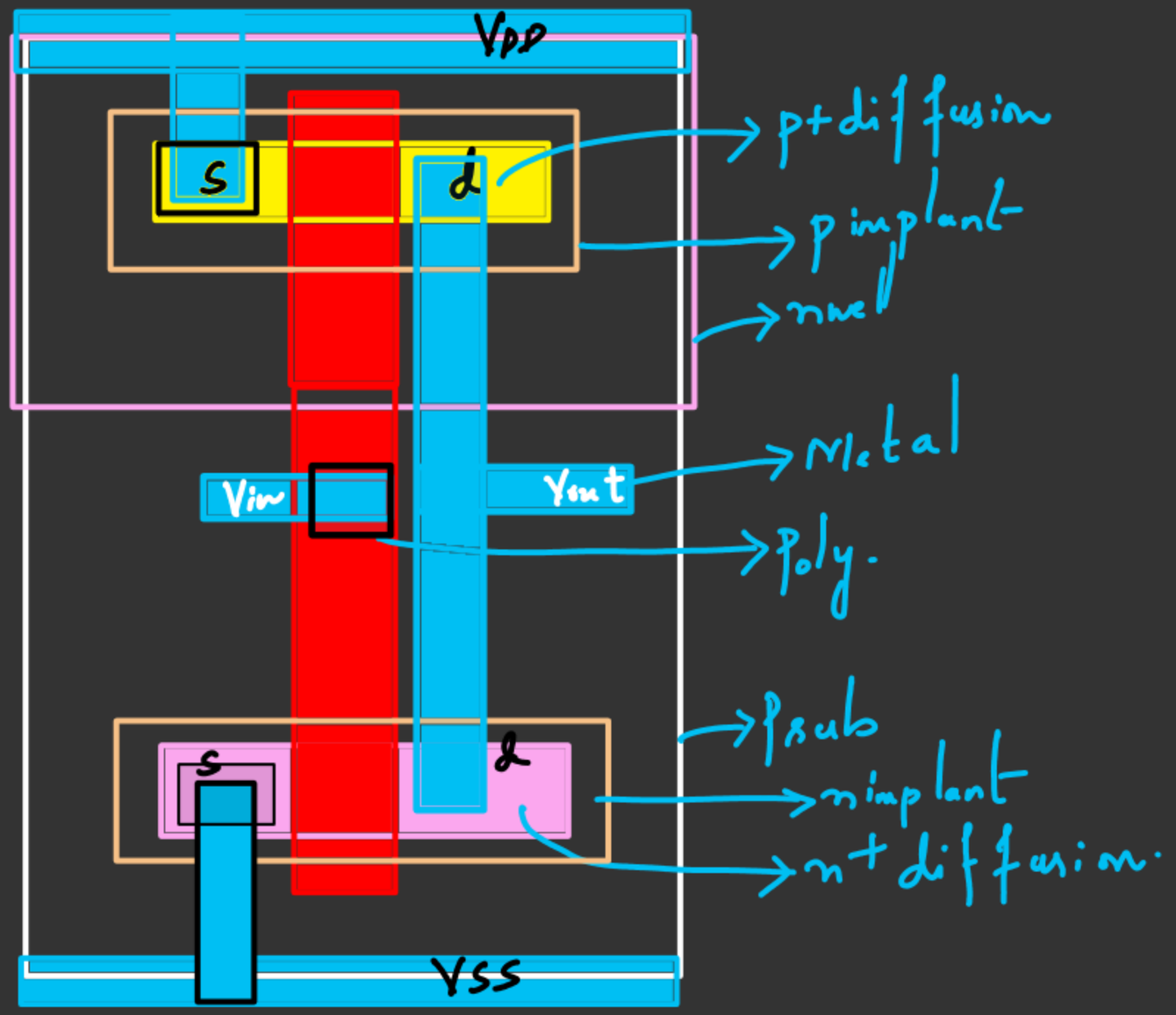
↳ GDSII → Graphical data stream information interchange.
↳ GDSII has Complete layout of standard cells

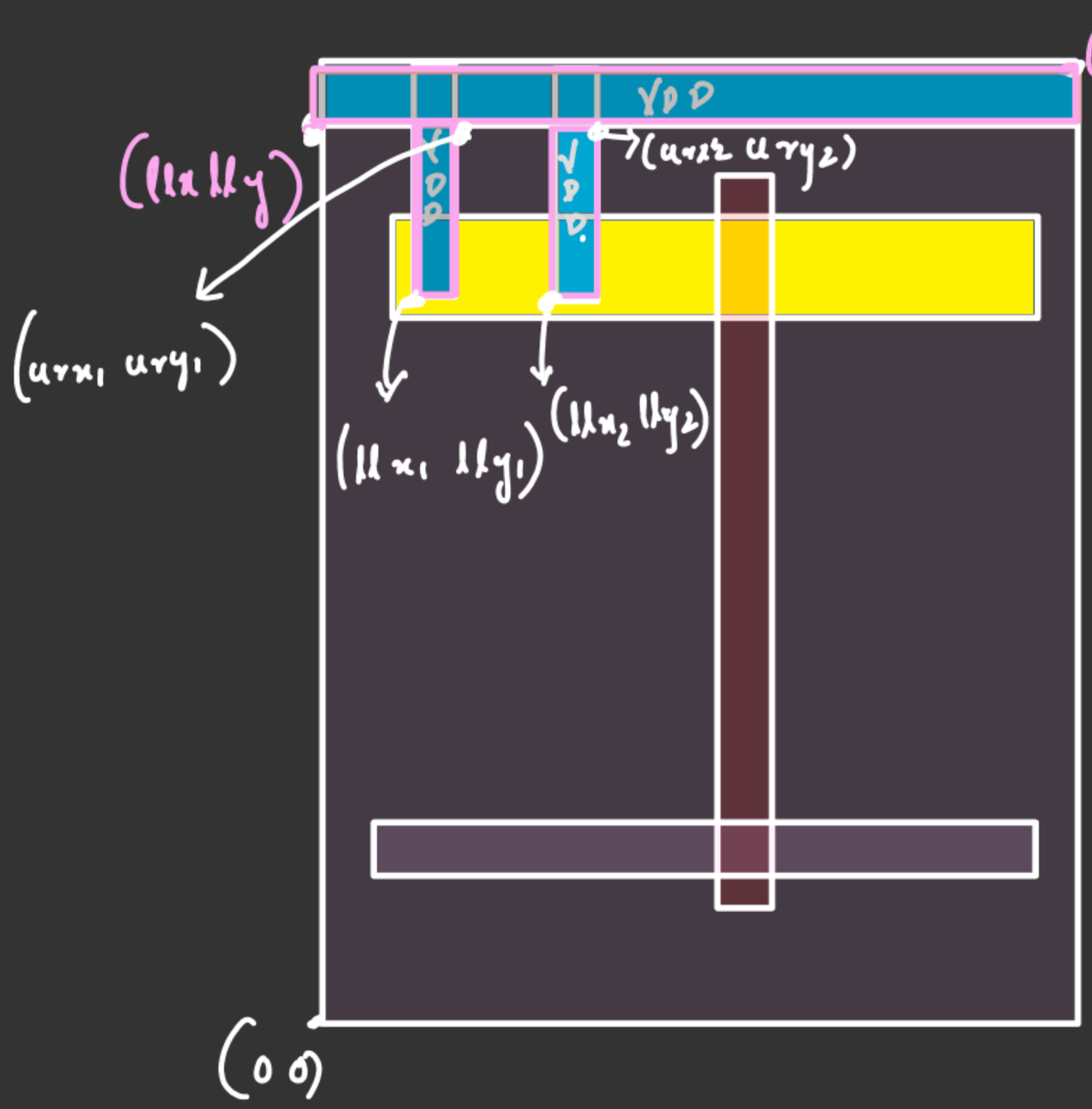
Content of LEF:

↳ Width and height of std_cell
↳ Allowed orientation of cells
↳ physical pin information of cell
↳ metal layer of pin
↳ Co-ordinates of pin layers
↳ obstruction blockage layer co-ordinates.



Layout of std.





bbox

(llx lly urx ury)

VDD -> M1

rect (llx lly urx ury)
rect (llx1 lly1 urx1 ury1)
rect (llx2 lly2 urx2 ury2)

④ tlut file.

↳ Resistance & Capacitance per micro meter for standard cells

↳ (max tlut → Rc Values will give more net delay

↳ (min tlut → Rc Values will give less net delay.

⑤ Netlist - (.v)

↳ ORCA_TOP.v

↳ 32bit RISC Core Processor